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DETAIL SPECIFICATION
MANUALS, TECHNICAL: SAMPLE BASIC WEIGHT CHECKLISTS AND LOADING DATA

Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be submitted to: ESC/HGGI, 4170 Hebble Creek Road, Bldg. 280, Door 15, Area A, Wright-Patterson AFB, OH 45433–5653 or by email to:SGMLsupport@wpafb.af.mil. Since contact information can change, the currency of this address information should be verified using the ASSIST Online database at https://assist.daps.dla.mil/.

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1 SCOPE

1.1 Scope. This detail specification covers the preparation of the Sample Basic Weight Checklists and Loading Data aircraft manuals. These manuals consist of TO 1X-XXXX-5-1, Sample Basic Weight Checklists (Chart A) and TO 1X-XXXX-5-2, Loading Data (Chart E) as approved for use in TO 1-1B-40 and development under provisions of SAWE-RP7. In addition to "paper" delivery, this specification provides for electronic delivery of data through the use of the Document Type Definitions (DTD) contained in Appendixes A and B.

1.2 Detail. The level of detail contained in this specification is necessary to comply with the requirements of the Joint Computer-aided Acquisition and Logistics Support (JCALS) system.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-38784	Manuals, Technical: General Style and Format Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

AIR FORCE TECHNICAL MANUALS

TO 00–5–3	AF Technical Order Life Cycle Management
TO 1-1B-40	Weight and Balance Data
TO 1-1B-50	Basic Technical Order for USAF Aircraft Weight and Balance

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the acquiring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

No non-government publications are cited as first tier requirements in this specification. See appendices for additional information.

SOCIETY OF ALLIED WEIGHT ENGINEERS

SAWE-RP7	Mass Properties Management and Control for Military Aircraft.

(Copies of SAWE recommended practices should be addressed to the Society of Allied Weight Engineers, P.O. Box 60024, Terminal Annex, Los Angeles, CA 90060 or http://www.sawe.org/.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 Source of technical data. Charts A and E shall be used as the contents of TO 1X-XXXX-5-1 (Sample Basic Weight Checklists) and TO 1X-XXXX-5-2 (Loading Data) manuals for the applicable Type/Model/Series aircraft or rotorcraft (see 6.2b.). The contractor shall prepare this data, using ESC/HGGI, for Air Force completion of the Weight and Balance Handbook. This data shall be for representative or individual aircraft as specified in TO 1-1B-50.

3.2 Development and preparation. The general manner of development and preparation for manuals shall be in accordance with the requirements of MIL-STD-38784.

3.3 Sample basic weight checklists (TO 1X-XXXX-5-1).

3.3.1 Sample basic weight checklists manual arrangement. The contractor shall use the Document Type Definition (DTD) that is available at the TMSS website (see A.4.1). If electronic delivery of this manual is required (see 6.2c.). Each sample Basic Weight Checklists manual shall consist of the following:

a. Front Matter

b. Chapter 1 - Introduction

c. Chapter 2 - Sample Basic Weight Checklists (Chart A)

3.3.1.1 Front matter. Unless otherwise specified by the acquiring activity (see 6.2d.), illustrations of the Chart A item locations shall be included. The front matter shall consist of a title page and table of contents in accordance with the requirements of MIL-STD-38784. The following is an exception to the general front matter requirements:

Electronic media requirement: The front matter shall include a list of changes.

Printed media requirement: The front matter shall include a list of effective pages. The sample Basic Weight Checklists and diagrams shall be legible, either when the manual is viewed vertically (normal reading position), or when the manual is rotated 90 degrees clockwise from the vertical position. Illustrations of the Chart A item locations shall be inserted as a facing page to the corresponding items listing (see figure 3.).

3.3.1.1.1 Sample basic weight checklists title. The words "SAMPLE BASIC WEIGHT CHECKLISTS" shall be used as the type of publication. The aircraft type designation shall be used as the prime title.

3.3.1.2 Chapter 1 - Introduction. Chapter 1 shall contain the foreword/preface/introduction requirements of MIL-STD-38784. In addition, it shall include an explanation of how to use and maintain the sample Chart A, and create a new Chart A (see figure 1).

3.3.1.3 Chapter 2 - Sample basic weight checklists (Chart A). Chapter 2 shall begin by identifying the sample Basic Weight Checklists contained in the manual (see figure 2). The remainder of Chapter 2 shall consist of the sample Basic Weight Checklists. These lists shall be identical to the final approved Chart A, as specified in 3.1, except that the words "Sample Only" shall be printed immediately following the aircraft Type/Model/Series designation in the Model/Design/Series block and the "RECORD OF CHECKING" columns shall be blank.

3.4 Loading data (TO 1X-XXXX-5-2). Print media requirement: Loading Data, charts, tables, and diagrams shall be legible either when the manual is viewed vertically (normal reading position), or when the manual is rotated 90 degrees clockwise from the vertical position.

3.4.1 Loading data manual arrangement. The contractor shall use the Document Type Definition (DTD) available at the TMSS website (see A.4.1). If electronic delivery of this manual is required (see 6.2c.). Each Loading Data manual shall consist of the following:

a. Front Matter

b. Chapter 1 - Introduction

c. Chapter 2 - Loading Data

3.4.1.1 Front matter. The front matter shall consist of a title page and table of contents in accordance with the requirements of MIL-STD-38784. The following is an exception to the general front matter requirements:

Electronic media requirement: The front matter shall include a list of changes.

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Printed media requirement: The front matter shall include a list of effective pages.

3.4.1.1.1 Loading data title. The words "LOADING DATA" shall be used as the type of publication. The aircraft type designation shall be used as the prime title.

3.4.1.2 Chapter 1 -Introduction. Chapter 1 shall contain the foreword/preface/introduction requirements of MIL-STD-38784. In addition, it shall include a weight and balance classification reference, and an explanation of how to use the Chart E (see figure 4).

3.4.1.3 Chapter 2 - Loading Data. Chapter 2 shall begin by delineating weight and balance control requirements, aircraft weighing requirements and intervals, and the contained Loading Data (Chart E). The weighing intervals shall be determined by the acquiring activity (see figures 5 and 6). The remainder of Chapter 2 shall consist of the Loading Data identical to the final approved Chart E as specified in 3.1. Chart E shall begin with general aircraft weighing instructions, aircraft diagram, and general notes affecting aircraft loading. The remaining Chart E content, such as loading tables, graphs, etc., shall appear in the order of use on the Form F - Weight and Balance Clearance Form (DD Form 365-4), as represented in figures 5, 6 and 7. Necessary Chart E additions or deletions shall be made as specified or approved by the acquiring activity (see 6.2e. and 6.2f.).

4 VERIFICATION

4.1 Verification requirements. When the technical data produced according to this specification is offered for acceptance, all tests, reviews, and verifications required by the acquiring activity to determine that it conforms to the requirements in section 3 of the specification, shall be performed as specified in the solicitation or contract. The Air Force Technical Order Policy and Procedures (AF TOPP) team, AFMC/A4UE, provides the specific requirements for verification of technical data developed and delivered through this specification, as well as guidance for including these requirements in the solicitation or contract (see TO 00-5-3, AF Technical Order Life Cycle Management, 2.2.2).

4.2 Compliance. Technical manuals (TMs) shall meet all requirements of section 3 of this specification and the appropriate DTD appendix, as required by the acquiring activity (see 6.2). The requirements set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies, submitted to the government for acceptance, comply with all requirements of the contract. Use of sampling inspections shall be at the discretion of the contractor, and in accordance with commercially acceptable quality assurance procedures. However, use of sampling in QA procedures does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the military system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The technical manuals prepared in accordance with this document provide required information to service activities for the preparation of checklists that will be inserted into the Weight and Balance Handbook for individual aircraft. The necessary Loading Data and restrictions required to complete DD Form 365-4 will also be provided. These manuals provide guidance and instruction for

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specific Type/Model/Series aircraft to better facilitate compliance with weight and balance requirements. In addition, the Sample Basic Weight Checklists and Loading Data manuals serve as the baseline for updating individual aircraft Weight and Balance Handbooks.

6.2 Acquisition requirement . Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. Charts A and E, as approved by the acquiring activity engineering personnel (see 3.1).

c. If electronic delivery of the manuals is required (see 3.3.1 and 3.4.1).

d. If illustrations of the Chart A item locations are to be other than as specified in this document (see 3.3.1.3).

e. The weighing intervals to be included (see 3.4.1.3).

f. If Chart E additions or deletions are to be made (see 3.4.1.3).

g. If representative aircraft are within + percent MAC CG and + pounds (see figure 5).

h. Address of logistics center assigned maintenance engineering responsibility for the specific type aircraft (see figures 5 and 6).

i. If specific weighing requirements are to be specified as subparagraphs (see figures 5 and 6).

j. Packaging requirements (see 5.1).

6.3 Acronyms. The acronym(s) used in this document are defined as follows:

AFI	Air Force Instruction
ASSIST	Acquisition Streamlining and Standardization Information System
CG	Center of Gravity
DD	Defense Department
DOD	Department of Defense
DTD	Document Type Definition
MAC	Mean Aerodynamic Chord
TCTO	Time Compliance Technical Order
TO	Technical Order

6.4 Definitions. To clarify the terms used throughout this specification, the following definitions are given:

6.4.1 Verification. Verification (section 4) in the context of this specification equates to the contractor's quality assurance program for validating the content of the manuals. Suggested validation methods include:

a. Actual performance. Using production configured equipment, hands-on performance of the procedure using the technical instructions as written.

b. Simulation. Using production configured equipment and the manual, simulate the actions required by the task steps.

c. Table top analysis. Primarily for nonprocedural data, compare the technical content to source data to ensure the technical accuracy and depth of coverage.

6.4.2 Mean aerodynamic chord . An engineering term which represents an airfoil's chord in aircraft design. It is a constant length which is used in the calculation of center of gravity locations in terms of percent of MAC.

6.5 Aircraft weight and balance classification. Aircraft weight and balance classifications (Class 1 and Class 2 aircraft) are defined in TO 1-1B-50 and SAWE-RP7,

6.6 Charts A and E. Chart A is the DD Form 365-1, Basic Weight Checklist Record. Chart E is not a standard form, but may be any of the graphs, charts, or tables that are contained in TO 1X-XXXX-5-2 that provides loading data.

6.7 Technical manual. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been authorized and assigned an Acquisition Management Systems Control (AMSC) number must be listed on a

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separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.8 Subject term (key word) listing.

Arm
Chart A
Chart E
Class 1 Aircraft
Class 2 Aircraft
DD Form 365
Form F
Moment
Weight and Balance
Percent MAC

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CHAPTER 1 INTRODUCTION

1.1 PURPOSE AND SCOPE.

This technical manual contains Sample Basic Weight Checklists (Charts A) which are applicable to model F-15 aircraft. These charts are intended to assist using personnel in complying with the requirements of TO 1-1B-50 by providing data for insertion into the Weight and Balance Handbook.

1.2 HOW TO USE SAMPLE CHARTS A.

The Sample Charts A are tabulations of all fixed operating equipment items which have definite locations, may be installed, or are alternate installations for standard equipment items in the aircraft. The weight, arm, and simplified moment are given for each Chart A item. The Sample Charts A presented herein are intended to be used only as a guide in preparing a new Basic Weight Checklist for insertion into the Weight and Balance Handbook for representative or individual aircraft. Do not use sample Chart A to inventory equipment on the aircraft. Detailed procedures for preparing and maintaining Basic Weight Checklists (Chart A) are contained in TO 1-1B-40.

1.3 CREATING NEW CHARTS A.

When it is necessary to create a new Chart A, use the enclosed Sample Basic Weight Checklist in the Weight and Balance Handbook by deleting or crossing out the words "SAMPLE ONLY" on each page, and adjusting the items to match the individual aircraft configuration. Do this by crossing out those items not applicable to the aircraft, and adding new items, as appropriate. When the Sample Basic Weight Checklists are converted to the individual aircraft configuration, maintain it as a checklist, not as part of TO 1X-XXXX-5-1.

1.4 ROUND OFF.

The weights and arms are rounded to whole numbers. Simplified moments are rounded to one decimal place. Use the arm for inventorying the aircraft, and the weight and moment for weight tracking. Because of the round off error, multiplying weight times arm may not reflect the published moment; likewise, a moment divided by weight may not reflect the published arm.

1.5 RECORD OF APPLICABLE TIME COMPLIANCE TECHNICAL ORDERS (TCTO).

The record of applicable time compliance technical orders is a list of all TCTOs which affect the technical content (text or illustrations) of this manual. The Change/Revision/Supplement Data column lists the date of issue when each change was (or will be) incorporated into this manual. Only currently effective changes are listed. A TCTO is deleted from the list when either the applicable equipment configuration is no longer covered in the publication, or it is rescinded, superseded, or replaced.

1-6 YOUR RESPONSIBILITY TO LET US KNOW.

Every effort is made to keep the manual current; however, we cannot correct an error unless we know of its existence. In this regard it is essential that you do your part. Comments, corrections, and questions regarding this manual, or any phase of the basic weight and loading data, are welcome. These should be forwarded on AF Form 847 as directed by AFI 11-215 through your command headquarters.

FIGURE 1. Example Chapter 1 Introduction – sample basic weight checklist manual.

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CHAPTER 2 SAMPLE BASIC WEIGHT CHECKLISTS (CHARTS A)

2.1 PURPOSE.

The Sample Basic Weight Checklists contained herein are to be used only as guides for preparing new Charts A for insertion into the Weight and Balance Handbook for representative or individual aircraft. Preparation procedures for Chart A are found in T.O. 1-1B-40.

2.2 SAMPLE CHART A, F-15A, 72-113 THRU 72-115 AND 72-119.

Figure 2-1 contains Chart A data which reflects an aircraft configuration which is representative of Air Force Serial Number 72-113 thru 72-115 and 72-119 aircraft. The physical location of each fixed equipment item is shown in the view on the facing page opposite the Chart A listing.

2.3 SAMPLE CHART A, F-15A 73-085 AND UP.

Figure 2-2 contains Chart A data which reflects an aircraft configuration which is representative of Air Force Serial Number 73-085 and up aircraft. The physical location of each fixed equipment item is shown in the view on the facing page opposite the Chart A listing.

2.4 SAMPLE CHART A, TF-15A 73-108 AND UP.

Figure 2-3 contains Chart A data which reflects a model TF-15A aircraft configuration and is representative of Air Force Serial Number 73-108 and up aircraft. The physical location of each fixed equipment item is shown in the view on the facing page opposite the Chart A listing.

FIGURE 2. Example chapter 2 lead-in, sample basic weight checklists manual.

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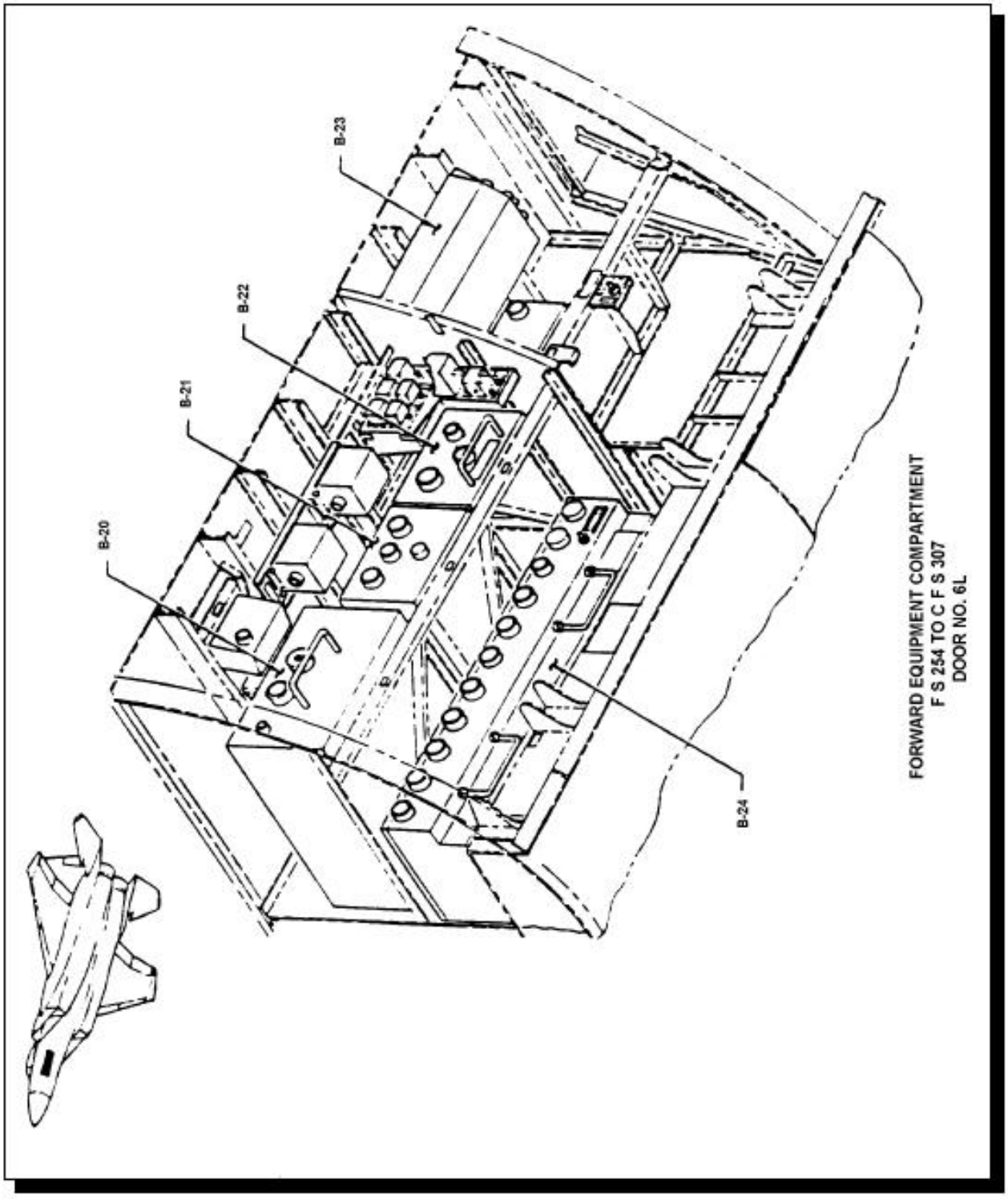


FIGURE 3. Example chapter 2, sample basic weight checklists - sample basic weight checklists manual.

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CHAPTER 1 INTRODUCTION

1.1 PURPOSE AND SCOPE

This technical manual contains Loading Data (Chart E) which are applicable to Model F-15A aircraft. These charts are intended to assist using personnel in complying with the requirements of TO 1-1B-50 by providing data for insertion into the DD365 series forms for *(representative/ individual) aircraft. This information is of a specific nature and, except where specifically stated herein, does not relieve any of the general requirements for USAF aircraft weight and balance found in TO 1-1B-50. *(Contractor, insert the correct term as applicable per TO 1-1B-50, Section IV.)

1.2 WEIGHT AND BALANCE CLASSIFICATION

In accordance with the criteria presented in TO 1-1B-50, the weight and balance classification of clearance will be as required by TO 1-1B-50 for Class *___ aircraft.

*NOTE

The contractor shall fill in the aircraft model and the weight and balance classification in accordance with that specified in Section IV of TO 1-1B-50. TO 1-1B-50 is the controlling document for USAF aircraft weight and balance classification. Recommendations for changing aircraft weight and balance classification will be made on AF Form 847 in accordance with AFI 11-215.

1.3 HOW TO USE CHART E

Chart E provides data necessary to comply with DD Form 365-4 (Form F) Weight and Balance Clearance requirements. Weight and simplified moments are obtained from Chart E for all the variable load items and are added, in the appropriate reference on Form F, to the aircraft's current basic weight and simplified moment from Chart C (DD Form 365-3). This total represents the gross weight and simplified moment of the loaded aircraft. In-flight center of gravity effects, such as fuel and bomb expenditures, are checked by subtracting the weights and simplified moments of such items from the takeoff condition. The resultant new weight and simplified moment are checked to ensure that the center of gravity remains within limits during the entire flight. Detailed instructions for preparing the DD Form 365-4 are contained in TO 1-1B-40.

1.4 RECORD OF APPLICABLE TIME COMPLIANCE TECHNICAL ORDERS (TCTOS)

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1.5 YOUR RESPONSIBILITY TO LET US KNOW

Every effort is made to keep the manual current; however, we cannot correct an error unless we know of its existence. In this regard it is essential that you do your part. Comments, corrections, and questions regarding this manual, or any phase of the basic weight and loading data are welcome. These should be forwarded on AF Form 847 as directed by AFI 11-215 through your command Headquarters.

FIGURE 3. Example chapter 2, sample basic weight checklists - sample basic weight checklists manual. - Continued.

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CHAPTER 1 INTRODUCTION

1.1 PURPOSE AND SCOPE

This technical manual contains Loading Data (Chart E) which are applicable to Model F-15A aircraft. These charts are intended to assist using personnel in complying with the requirements of TO 1-1B-50 by providing data for insertion into the DD365 series forms for *(representative/ individual) aircraft. This information is of a specific nature and, except where specifically stated herein, does not relieve any of the general requirements for USAF aircraft weight and balance found in TO 1-1B-50. *(Contractor, insert the correct term as applicable per TO 1-1B-50, Section IV.)

1.2 WEIGHT AND BALANCE CLASSIFICATION

In accordance with the criteria presented in TO 1-1B-50, the weight and balance classification of clearance will be as required by TO 1-1B-50 for Class *___ aircraft.

*NOTE

The contractor shall fill in the aircraft model and the weight and balance classification in accordance with that specified in Section IV of TO 1-1B-50. TO 1-1B-50 is the controlling document for USAF aircraft weight and balance classification. Recommendations for changing aircraft weight and balance classification will be made on AF Form 847 in accordance with AFI 11-215.

1.3 HOW TO USE CHART E

Chart E provides data necessary to comply with DD Form 365-4 (Form F) Weight and Balance Clearance requirements. Weight and simplified moments are obtained from Chart E for all the variable load items and are added, in the appropriate reference on Form F, to the aircraft's current basic weight and simplified moment from Chart C (DD Form 365-3). This total represents the gross weight and simplified moment of the loaded aircraft. In-flight center of gravity effects, such as fuel and bomb expenditures, are checked by subtracting the weights and simplified moments of such items from the takeoff condition. The resultant new weight and simplified moment are checked to ensure that the center of gravity remains within limits during the entire flight. Detailed instructions for preparing the DD Form 365-4 are contained in TO 1-1B-40.

1.4 RECORD OF APPLICABLE TIME COMPLIANCE TECHNICAL ORDERS (TCTOS)

The record of applicable time compliance technical orders is a list of all TCTOs which affect the technical content (text or illustrations) of this manual. The Change/Revision/Supplement Data column lists the date of issue when each change was (or will be) incorporated into this manual. Only current TCTOs are listed. A TCTO is deleted from the list when either the applicable equipment configuration is no longer covered in the publication, or it is rescinded, superseded, or replaced.

1.5 YOUR RESPONSIBILITY TO LET US KNOW

Every effort is made to keep the manual current; however, we cannot correct an error unless we know of its existence. In this regard it is essential that you do your part. Comments, corrections, and questions regarding this manual, or any phase of the basic weight and loading data are welcome. These should be forwarded on AF Form 847 as directed by AFI 11-215 through your command Headquarters.

FIGURE 4. Example chapter 1, introduction - loading data.

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CHAPTER 2 LOADING DATA

2.1 WEIGHT AND BALANCE REQUIREMENTS.

2.1.1 General. This chapter provides specific requirements for weight and balance control of Model T-37B aircraft in accordance with the general requirements found in TO 1-1B-50. This information is of a specific nature and, except where specifically stated herein, does not relieve any of the general requirements of TO 1-1B-50.

2.1.1.1 Maintaining Manuals. Weight and Balance personnel (per TO 1-1B-50) will maintain a current Weight and Balance Handbook for each representative aircraft.

NOTE

A representative aircraft is one which serves to represent all aircraft that are within ± ____* percent MAC center of gravity and ± ____* pounds of the representative aircraft basic weight center of gravity location and basic weight. To satisfy these criteria, more than one representative aircraft may be required for proper weight and balance monitoring of all assigned aircraft. *(Shall be determined by acquiring activity, see 6.2h.)

2.1.1.2 Specific Requirements. Specific weight and balance requirements will be listed here as subparagraphs.

2.1.2 Weighing Requirements. In addition to the general weighing requirements specified in Section IV of TO 1-1B-50, T-37B model aircraft will be inventoried and weighed at least once every 48 months. This time interval weighing is required to ensure that the cumulative effects, of minor modifications and repairs, on the aircraft basic weight and center of gravity location, are accurately known. Aircraft time interval weighing should be scheduled in conjunction with the scheduled phase inspection which will precede the 48-month time interval limit.

2.1.3 Weighing Record. After completion of the Aircraft Weighing Record (DD Form 365-2), forward one copy to:

List the address of the Logistics Center which is assigned the maintenance engineering management responsibility for the aircraft (see 6.2i).

2.1.4 Chart E, T-37B 59-256 and up. Figure 2-1 contains the Chart E Loading Data, applicable to model T-37B aircraft Serial Numbers 59-256 and up, which are necessary for computing aircraft weight and balance, and for completion of DD Form 365-4 Weight and Balance Clearance Form F, when required.

FIGURE 5. Example chapter 2 lead-in, loading data manual (class 1 aircraft).

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CHAPTER 2 LOADING DATA

2.1 WEIGHT AND BALANCE REQUIREMENTS

2.1.1 General. This chapter provides specific requirements for weight and balance control of Model F/TF-15A aircraft in accordance with the general requirements found in TO 1-1B-50. This information is of a specific nature and, except where specifically stated herein, does not relieve any of the general requirements of TO 1-1B-50.

2.1.1.1 Maintaining Manuals. Weight and balance personnel (per TO 1-1B-50) will maintain a current Weight and Balance Handbook for each aircraft.

2.1.1.2 Center of Gravity. It is possible to exceed the center of gravity limits of the aircraft with certain store configurations. Therefore, weight and balance clearance (DD Form 365-4), in accordance with Section IV of TO 1-1B-50, will be accomplished prior to each flight.

2.1.1.3 Specific Requirements. Specific weight and balance requirements will be listed here as subparagraph.

2.1.2 Weighing Requirements. In addition to the general weighing requirements specified in Section IV of TO 1-1B-50, F/TF-15A model aircraft will be inventoried and weighed at least once every 36 months. This time interval weighing is required to ensure that the cumulative effects, of minor modifications and repairs, on the aircraft basic weight and center of gravity location, are accurately known. Aircraft time interval weighing should be scheduled in conjunction with the scheduled phase inspection which will precede the 36-month time interval limit.

2.1.2.1 Specific Weighing Requirements. When required by the acquiring activity, specific weighing requirements such as after engine change, etcetera, will be specified here as subparagraphs (see 6.2j).

2.1.3 Weighing Record. After completion of the Aircraft Weighing Record (DD Form 365-2), forward one copy to:

List the address of the Air Logistics Center which is assigned the maintenance engineering management responsibility for the aircraft (see 6.2i).

2.1.4 Chart E, F/TF-15A 72-113 and up. Figure 2-1 contains the Chart E Loading Data, applicable to model F/TF-15A aircraft Serial Numbers 72-113 and up, which are necessary for computing aircraft weight and balance, and for completion of DD Form 365-4 Weight and Balance Clearance Form F, when required.

FIGURE 6. Example chapter 2 lead-in, loading data manual (class 2 aircraft).

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Weight and Balance Requirements
Weighing Requirements
Weighing Record
General Weighing Instructions
    Aircraft Conditions
    Fuel Drains
    Engine Oil
    Procedures
    Leveling
    Measuring
Aircraft Diagram
  * Jack Points
    Dimensions Pertaining to Aircraft Weight and Balance
    Fuel Tank Arrangement
Notes Affecting Aircraft Loading
    Engine Oil Table
    Crew Tables
    Ammunition Tables
    Stores Tables and Diagrams
    Fuel Tables
    Water Injection Fluid Tables
    Miscellaneous Data Tables
    Center of Gravity Limits
    Instructions for Loading
        Sample Form F
        Forward CG Calculations
        Aft CG Calculations
    Typical Service Load Conditions
    Takeoff and Landing Gross Weight Restrictions
    Center of Gravity and Percent MAC Calculations
    Moment/Weight CG Conversion Table
    Gross Weight vs. CG Position Plots

* Identify if nose or tail jack points were used, and which points were used to calculate weight and CG position.

FIGURE 7. Example order for chart E data loading data manual.

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APPENDIX A

SAMPLE BASIC WEIGHT CHECKLISTS DOCUMENT TYPE DEFINITION (DTD) SUBSET

A.1 SCOPE.

A.1.1 Scope. DTDs are based on rules outlined in MIL-PRF-28001, and ISO 8879. The DTD subset referenced within this appendix provides the structure and content of documents prepared in accordance with this detail specification. For digital copies of the DTD, Tag Description Table, and the Attribute Description Table, see A.4.1. This appendix is a mandatory part of this detail specification. The information herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS.

A.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

A.2.2 Goverment document.

A.2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, issues of these documents are listed in the ASSIST data base (https://assist.daps.dla.mil) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-28001 Markup Requirements and Generic Style Specification for Electronic Printed Output and Exchange of Text

A.2.3 Non-government publications.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 8879 (DoD adopted) Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)

(Copies of these documents are available online at https://assist.daps.dla.milor from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

A.3 DIGITAL SUPPORT SUITES.

The following comprise the tools contained in the Digital Support Suite (DSS) used for authoring and rendering the sample basic weight checklists doc type.

A.3.1 Document Type Definition. Data to be delivered digitally in accordance with this specification shall be tagged using the sample basic weight checklists DTD found at the TMSS website (see A.4.1). This DTD provides the structure and content in accordance with the detail specification. The procedure for accomplishing this is found in MIL-PRF-28001 and ISO 8879.

A.3.2 Tag Description Table. The Tag Description Table (TDT) provides detailed descriptions of the elements contained in the sample basic weight checklists DTD. The TDT contains the following: the element tagging structure, full element name, tag minimization requirements, element structure, reference elements, source paragraph, and attribute descriptions unique to the element. See A.4.1 for information on obtaining this file in digital format.

A.3.3 Attribute Description Table. The Attribute Description Table (ATD) provides detailed descriptions of the attributes contained in the sample basic weight checklists DTD. See A.4.1 for information on obtaining this file in digital format.

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APPENDIX A

A.4 OBTAINING DSS FILES.

A.4.1 Obtaining DSS files with a .mil address . DTDs, TDTs, and other files in the DSS can be accessed at the Technical Manual Specifications and Standards (TMSS) website https://techdata.wpafb.af.mil/tmss/index.html. On the web page, the "Baseline" menu option in the left pane contains three bulleted options called "Specifications", "Standards", and "Handbooks". Hover the cursor over "Specifications" and a listing of the TMSS specifications will appear. Hover over the desired specification number and another drop down list will appear that shows a list containing a PDF version of the specification and the associated appendices. To obtain the preferred DTD, select the desired appendix. The following items will appear on the downloading page: The name of the specification, the appendix number and name, the current version of the DSS, buttons to download the DTD, FOSI, screen FOSI, ATD, and the TDT, and a "Download" button to download the entire DSS zip file.

A.4.2 Obtaining files without a .mil address . For addresses other than .mil, access to the TMSS Community of Practice (CoP) must be attained via the Air Force Portal. The Air Force Portal help desk can be reached commercially at (334) 416-5771 option 7 or toll free at (877) 596-5771 option 7. Once access is attained, navigate to the TMSS CoP at https://afkm.wpafb.af.mil/community/views/home.aspx?Filter=OO-LG-MC-38 to download DTDs, TDTs, and other files contained in the DSSs. A DSS can be downloaded by selecting the "Digital Support Suites (DSSs)" link under the "Document Management" section. A listing of the most current zip files for the DSSs appears. Select the appropriate zip file to download.

A.4.3 TMSS Support. Please address any comments or questions relating to the DSS and/or specifications and standards to SGMLSUPPORT@wpafb.af.mil or Technical Manual Specifications and Standards ESC/HGGI 4170 Hebble Creek Road, Building 280, Door 15, Wright-Patterson AFB OH 45433-5006.

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APPENDIX B

LOADING DATA MANUAL DOCUMENT TYPE DEFINITION SUBSET

B.1 SCOPE

B.1.1 Scope. DTDs are based on rules outlined in MIL-PRF-28001, and ISO 8879. The DTD subset referenced within this appendix provides the structure and content of documents prepared in accordance with this detail specification. For digital copies of the DTD, Tag Description Table, and the Attribute Description Table, see A.4.1. This appendix is a mandatory part of this detail specification. The information herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS.

B.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

B.2.2 Government document.

B.2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, issues of these documents are listed in the ASSIST data base (https://assist.daps.dla.mil) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-28001 Markup Requirements and Generic Style Specification for Electronic Printed Output and Exchange of Text

B.2.3 Non-government publications.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 8879 (DoD adopted) Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)

(Copies of these documents are available online at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

B.3 DIGITAL SUPPORT SUITES.

The following comprise the tools contained in the Digital Support Suite (DSS) used for authoring and rendering the loading data doctype.

B.3.1 Document Type Definition. Data to be delivered digitally in accordance with this specification shall be tagged using the loading data doctype DTD found at the TMSS website (see A.4.1). This DTD provides the structure and content in accordance with the detail specification. The procedure for accomplishing this is found in MIL-PRF-28001 and ISO 8879.

B.3.2 Tag Description Table. The Tag Description Table (TDT) provides detailed descriptions of the elements contained in the loading data doctype DTD. The TDT contains the following: the element tagging structure, full element name, tag minimization requirements, element structure, reference elements, source paragraph, and attribute descriptions unique to the element. See A.4.1 for information on obtaining this file in digital format.

B.3.3 Attribute Description Table. The Attribute Description Table (ATD) provides detailed descriptions of the attributes contained in the loading data doctype DTD. See A.4.1 for information on obtaining this file in digital format.

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APPENDIX B

B.4 OBTAINING DSS FILES.

B.4.1 Obtaining DSS files with a .mil address . DTDs, TDTs, and other files in the DSS can be accessed at the Technical Manual Specifications and Standards (TMSS) website https://techdata.wpafb.af.mil/tmss/index.html. On the web page, the "Baseline" menu option in the left pane contains three bulleted options called "Specifications", "Standards", and "Handbooks". Hover the cursor over "Specifications" and a listing of the TMSS specifications will appear. Hover over the desired specification number and another drop down list will appear that shows a list containing a PDF version of the specification and the associated appendices. To obtain the preferred DTD, select the desired appendix. The following items will appear on the downloading page: The name of the specification, the appendix number and name, the current version of the DSS, buttons to download the DTD, FOSI, Screen FOSI, ATD, and the TDT, and a "Download" button to download the entire DSS zip file.

B.4.2 Obtaining files without a .mil address . For addresses other than .mil, access to the TMSS Community of Practice (CoP) must be attained via the Air Force Portal. The Air Force Portal help desk can be reached commercially at (334) 416-5771 option 7 or toll free at (877) 596-5771 option 7. Once access is attained, navigate to the TMSS CoP at https://afkm.wpafb.af.mil/community/views/home.aspx?Filter=OO-LG-MC-38 to download DTDs, TDTs, and other files contained in the DSSs. A DSS can be downloaded by selecting the "Digital Support Suites (DSSs)" link under the "Document Management" section. A listing of the most current zip files for the DSSs appears. Select the appropriate zip file to download.

B.4.3 TMSS Support. Please address any comments or questions relating to the DSS and/or specifications and standards to SGMLSUPPORT@wpafb.af.mil or Technical Manual Specifications and Standards ESC/HGGI 4170 Hebble Creek Road, Building 280, Door 15, Wright-Patterson AFB OH 45433-5006.

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NOTE

The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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